Modeling with Partial Differential Equations
Exercises - Session 3: the Heat Equation

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Exercise 1 — Analytical solution of the one-dimensional heat equation We aim to solve the Cauchy problem

$$\begin{cases} \frac{\partial u}{\partial t}(t,x) = k\frac{\partial^2 u}{\partial^2 x}(t,x), & t>0, \quad x\in(0,L), \\ u(0,x) = u_0(x), & x\in(0,L), \\ u(t,0) = u(t,L) = 0, & t>0, \end{cases} \tag{1}$$

where $k>0$ is the thermal diffusion coefficient and $L>0$ the length of the medium in which heat diffuses. To this end, we use the method of separation of variables.

1. **Separation of variables**
   Seek a solution of the form $u(t,x) = T(t)X(x)$, for an initial condition u_0 which is not zero. What does the partial differential equation become when u is replaced by this product? Show that there exists $\lambda\in\mathbb{R}$ such that
   $$X''(x) + \lambda X(x) = 0$$
   and
   $$T'(t) + k\lambda T(t) = 0.$$

   Solution :

   (a) We seek solutions of the form:
   $$u(t,x) = T(t)X(x)$$
   where T is a function of time only and X is a function of space only. This assumption is called "separating variables" as it converts the PDE into ODEs for functions of single variables.
   We substitute $u(t,x) = T(t)X(x)$ into $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$ and compute the derivatives:
   $$\frac{\partial u}{\partial t}(t,x) = T'(t)X(x), \quad \frac{\partial^2 u}{\partial x^2}(t,x) = T(t)X''(x)$$
   We plug back into the PDE and we get for all $t>0$ and $x\in(0,L)$.
   $$T'(t)X(x) = kT(t)X''(x).$$
   Since u_0 is not zero, $T(0)\neq 0$ and there exists $x_0\in(0,1)$ such that $X(x_0)\neq 0$. By considering the previous equation at point $x=x_0$, one gets
   $$T'(t) = -k\lambda T(t)$$
   with $\lambda = -X''(x_0)/X(x_0)$. By then considering this equation at $t=0$, one gets
   $$\lambda = -T'(0)/(kT(0))$$
   and
   $$X''(x) = -\lambda X(x).$$

2. **Equation in x (Sturm–Liouville)**
The spatial solution $X(\cdot)$ must satisfy, for some λ,

$$X''(x) + \lambda X(x) = 0, \qquad X(0) = 0, \quad X(L) = 0. \tag{2}$$

For which values of λ (eigenvalues) does (2) admit nontrivial solutions $X(\cdot)$? What are those solutions (eigenfunctions)?

Solution : *After separation, the spatial equation reads*

$$X''(x) + \lambda X(x) = 0 \tag{3}$$

for $x \in (0, L)$, subject to homogeneous Dirichlet boundary conditions

$$X(0) = 0, \qquad X(L) = 0. \tag{4}$$

Inded, the conditions $u(t, 0) = u(t, L) = 0$ rewrite $T(t)X(0) = T(t)X(L) = 0$.

Since we search for non trivial solutions where $T(.)$ is not identically zero, it follows that $X(0) = X(L) = 0$. These conditions ensure the temperature is fixed at both ends.
Such a problem is called a Sturm–Liouville boundary value problem.

We consider problem (3)-(4), where λ is the separation constant.

We distinguish three cases based on the value of λ:

(a) ***Case 1:*** $\lambda = 0$
The ODE (3) reduces to

$$X''(x) = 0.$$

Integrating twice yields

$$X(x) = Ax + B,$$

where $A, B \in \mathbb{R}$ are constants.
Apply boundary conditions, we deduce

$$X(0) = B = 0, \quad X(L) = AL = 0.$$

To satisfy both, $B = 0$ and $A = 0$. Hence,

$$X(x) \equiv 0,$$

which is the trivial solution. No nontrivial eigenfunction exists for $\lambda = 0$.

(b) ***Case 2:*** $\lambda = -\beta^2 < 0$ ***for some*** $\beta > 0$
The ODE (3) rewrites to

$$X''(x) - \beta^2 X(x) = 0.$$

The general solution is

$$X(x) = Ae^{\beta x} + Be^{-\beta x},$$

or equivalently,

$$X(x) = C\cosh(\beta x) + D\sinh(\beta x),$$

with constants $C, D \in \mathbb{R}$.
Apply boundary conditions implies

$$X(0) = C = 0,$$

so

$$X(x) = D\sinh(\beta x).$$

In particular

$$X(L) = D\sinh(\beta L) = 0.$$

Since $\sinh(\beta L) \neq 0$ for $\beta L > 0$, we have $D = 0$.
Thus,

$$X(x) \equiv 0,$$

again the trivial solution. No nontrivial eigenfunction for $\lambda < 0$.

(c) ***Case 3:*** $\lambda = \omega^2 > 0$ ***for some*** $\omega > 0$

The ODE (3) *rewrites to*

$$X''(x) + \omega^2 X(x) = 0.$$

The general solution is

$$X(x) = A\cos(\omega x) + B\sin(\omega x).$$

Apply $X(0) = 0$ *leads to*

$$A = 0.$$

Therefore

$$X(x) = B\sin(\omega x),$$

ans in particular

$$X(L) = B\sin(\omega L) = 0.$$

For a nontrivial solution $B \neq 0$*, we require*

$$\sin(\omega L) = 0 \implies \omega L = n\pi, \quad n \in \mathbb{N}^*.$$

Hence,

$$\omega_n = \frac{n\pi}{L}, \quad \lambda_n = \omega_n^2 = \left(\frac{n\pi}{L}\right)^2.$$

The eigenfunctions corresponding to these eigenvalues are

$$X_n(x) = B_n \sin\left(\frac{n\pi x}{L}\right).$$

We conclude that the eigenvalues and eigenfunctions are respectively given by

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad X_n(x) = B_n \sin\left(\frac{n\pi x}{L}\right), \quad n \in \mathbb{N}^*.$$

These eigenfunctions form an orthogonal basis in $L^2((0, L))$ *and are fundamental for representing solutions of the heat equation via Fourier series expansions.*

3. **Equation in** t

For an eigenvalue λ_n, give the differential equation satisfied by $T_n(t)$. What is the associated family of temporal solutions $T_n(\cdot)$? Deduce a family of solutions to (1) for well chosen initial conditions u_0.

Solution : *For each eigen value* λ_n*, we have*

$$T_n'(t) + k\lambda_n T_n(t) = 0$$

This is a first-order linear ODE with constant coefficients. The solution, obtained by separation of variables

$$\frac{dT_n}{dt} = -k\lambda_n T_n \implies \frac{dT_n}{T_n} = -k\lambda_n dt.$$

and integration

$$\int \frac{dT_n}{T_n} = -k\lambda_n \int dt \implies \ln|T_n| = -k\lambda_n t + C.$$

is given by

$$T_n(t) = C_n e^{-k\lambda_n t},$$

where $C_n = e^C$ *is a constant determined by initial conditions. We thus have obtained a family of solutions* $u_n(t, x) = C_n e^{-k\lambda_n t}\sin\left(\frac{n\pi x}{L}\right)$ *indexed by* $n \in \mathbb{N}^*$*, for the initial conditions* $u_n(0, x) = C_n \sin\left(\frac{n\pi x}{L}\right)$*.*

Because $k > 0$ *and* $\lambda_n > 0$*, the solution decays exponentially as* $t \to \infty$*. Notice that this is consistent with the physical principle of heat diffusion dissipating temperature variations over time.*

4. **Finite sum solution**
Suppose $u_0(x)$ can be approximated or given exactly by a finite sum of sines

$$u_0(x) = \sum_{n=1}^{N} u_{0,n} \sin\left(\frac{n\pi x}{L}\right),$$

with a fixed integer N, where $(u_{0,1}, \ldots, u_{0,N}) \in \mathbb{R}^N$. Write the general solution $u(t,x)$ of the problem as a sum of sine functions that satisfy the initial condition.

Remark: Although we only work with finite sums here, one can also make sense of the solution when the sum becomes infinite, leading to what is called a Fourier series.

Solution :

Recall that each separable solution associated with eigenvalue λ_n has the form

$$u_n(t,x) = T_n(t)X_n(x) = A_n e^{-k\lambda_n t} \sin\left(\frac{n\pi x}{L}\right), \quad A_n = B_n C_n.$$

Because the heat equation is linear, the sum of any finite number of solutions is also a solution. By taking a finite sum of these eigenmodes, we can build al solution as

$$u(t,x) = \sum_{n=1}^{N} A_n e^{-k\lambda_n t} \sin\left(\frac{n\pi x}{L}\right).$$

The initial condition at time $t = 0$ becomes

$$u(0,x) = u_0(x) = \sum_{n=1}^{N} A_n \sin\left(\frac{n\pi x}{L}\right).$$

Since $u_0(x)$ is given by

$$u_0(x) = \sum_{n=1}^{N} u_{0,n} \sin\left(\frac{n\pi x}{L}\right),$$

with a fixed integer N, then the coefficients A_n are directly those $u_{0,n}$:

$$A_n = u_{0,n}, \quad \text{for } n = 1, \ldots, N,$$

and $A_n = 0$ for all $n > N$. The solution can thus be written as the finite sum:

$$u(t,x) = \sum_{n=1}^{N} u_{0,n} e^{-k\lambda_n t} \sin\left(\frac{n\pi x}{L}\right).$$

It involves only finitely many terms and explicitly shows how each sine mode decays in time.

Exercise 2 — Uniqueness of solutions to the heat equation Let $T > 0$, $k > 0$, and $f \in C^0([0,T]\times[0,L])$ be fixed. We aim to show that for any initial datum $u_0 \in C^0([0,L])$, there exists at most one solution $u \in C^{1,2}([0,T] \times [0,L])$ to the problem

$$\begin{cases} \partial_t u(t,x) = k\partial_{x,x} u(t,x) + f(t,x), & (t,x) \in (0,T) \times (0,L), \\ u(0,x) = u_0(x), & x \in (0,L), \\ u(t,0) = u(t,L) = 0, & t \in (0,T). \end{cases} \tag{5}$$

1. Using the linearity of the heat equation, show that uniqueness for (5) is equivalent to the statement that the zero solution $w \equiv 0$ is the only solution to the homogeneous problem:

$$\begin{cases} \partial_t w(t,x) = k\partial_{x,x} w(t,x), & (t,x) \in (0,T) \times (0,L), \\ w(0,x) = 0, & x \in (0,L), \\ w(t,0) = w(t,L) = 0, & t \in (0,T). \end{cases} \tag{6}$$

Solution : *Let u_1, u_2 be two solutions of (5) with the same data (u_0, f). By linearity,*

$$w := u_1 - u_2 \quad \text{satisfies} \quad \partial_t w = k\partial_{x,x} w, \quad w|_{t=0} = 0, \quad w|_{x=0} = w|_{x=L} = 0,$$

i.e. w solves (6). Hence, if $w = 0$ is the unique solution to (6), one gets $u_1 = u_2$. Conversely, if (6) admits a non-trivial solution w, then uniqueness does not hold for (5): indeed, for $f = u_0 = 0$, there are two different solutions, namely 0 and w. More generally, in this case, if u is a solution to (5) for some data (u_0, f), then $u + w$ ia also a solution to (5) for the same data (u_0, f)

2. Set $E(t) = \frac{1}{2}\int_0^L w^2(t,x)\,dx$ for all $t \in [0,T]$, where $w \in C^{1,2}([0,T]\times[0,L])$ is a solution of (6). Show that $E \in C^1([0,T])$ and that

$$\forall t \in (0,T), \quad E'(t) = \int_0^L w(t,x)\,\partial_t w(t,x)\,dx. \tag{7}$$

Solution : *Since $w \in C^{1,2}([0,T]\times[0,L])$, both w and $\partial_t w$ are continuous and bounded. Differentiating under the integral sign (Leibniz rule / dominated convergence) yields*

$$E'(t) = \frac{1}{2}\int_0^L \partial_t(w^2)(t,x)\,dx = \int_0^L w(t,x)\,\partial_t w(t,x)\,dx,$$

and $E \in C^1([0,T])$.

3. Show the following identity

$$\forall t \in [0,T], \quad E'(t) = -k\int_0^L \left(\partial_x w(t,x)\right)^2 dx, \tag{8}$$

and then conclude.

Solution : *Using $\partial_t w = k\partial_{x,x} w$ in the expression from (2),*

$$E'(t) = \int_0^L w\,\partial_t w\,dx = k\int_0^L w\,\partial_{x,x} w\,dx.$$

Integrate by parts in x:

$$\int_0^L w\,\partial_{x,x} w\,dx = \left[w\,\partial_x w\right]_0^L - \int_0^L (\partial_x w)^2\,dx.$$

Since $w(t,0) = w(t,L) = 0$ for all t, the boundary term vanishes, hence

$$E'(t) = -k\int_0^L (\partial_x w(t,x))^2\,dx \le 0.$$

In particular, E is nonincreasing and $E(t) \ge 0$ for all t. As $w(0,\cdot) = 0$, we have $E(0) = 0$. Thus $E(t) = 0$ for all $t \in [0,T]$. Therefore $\int_0^L (\partial_x w)^2\,dx = 0$ for all $t \in [0,T]$, so $\partial_x w(t,\cdot) \equiv 0$. Thus $w(t,\cdot)^2$ is continuous nonnegative and its integral is null, hence we have $w(t,x) = 0$ for all $(t,x) \in [0,T]\times[0,L]$. This gives uniqueness for (5).

4. Return to equation (5), and assume $g = 0$.
Let us admit the Poincaré inequality: for all $h \in C^1([0,L])$ such that $h(0) = h(L) = 0$, we have

$$\int_0^L h(x)^2\,dx \le \frac{L^2}{\pi^2}\int_0^L \left(h'(x)\right)^2 dx. \tag{9}$$

Using the same reasoning as in the previous question, show that

$$\forall t \in [0,T], \quad \int_0^L u(t,x)^2\,dx \le e^{-\frac{2k\pi^2}{L^2}t}\int_0^L u_0^2(x)\,dx \tag{10}$$

and deduce the convergence rate of the solution.

Solution : *For u solving $\partial_t u = k\partial_{x,x} u$ with $u|_{x=0,L} = 0$, define $E_u(t) = \frac{1}{2}\int_0^L u(t,x)^2\,dx$. As in (2)–(3),*

$$E_u'(t) = \int_0^L u\,\partial_t u\,dx = k\int_0^L u\,\partial_{x,x} u\,dx = -k\int_0^L (\partial_x u)^2\,dx.$$

By Poincaré's inequality,

$$\int_0^L (\partial_x u)^2\,dx \;\geq\; \frac{\pi^2}{L^2}\int_0^L u^2\,dx \;=\; \frac{2\pi^2}{L^2}E_u(t).$$

Therefore

$$E_u'(t) \leq -\frac{2k\pi^2}{L^2}\,E_u(t).$$

Grönwall's inequality yields

$$\forall t\in[0,T],\quad E_u(t)\leq E_u(0)\,e^{-\frac{2k\pi^2}{L^2}t},$$

which is the desired result since $E_u(0)=\frac{1}{2}\int_0^L u_0^2(x)\,dx$.

Remark*: All the results presented here also holds assuming that we consider solutions* $u\in C([0,T]\times[0,L])\cap C^{1,2}((0,T)\times(0,L))$*. Then* $E\in C([0,T])\cap C^1((0,T))$.

Exercise 3 - A proof of the one dimensional Poincaré's inequality We recall that $C_c^1((0,1))$ stands for the set of all continuously differentiable functions with compact support in the open interval $(0,1)$.

Theorem 0.1 (One-Dimensional Poincaré Inequality)**.** *Let* $\phi\in C_c^1((0,1))$*. Then the following inequality holds:*

$$\int_0^1 |\phi'(x)|^2dx \geq \pi^2\int_0^1 |\phi(x)|^2dx. \tag{11}$$

This inequality is fundamental in the calculus of variations, analysis, and partial differential equations. In this exercise, you will prove it by following an elegant argument that connects it to a simple variational problem. This proof has been given by Georgios Psaradakis in [1].

Consider a given test function $\phi\in C_c^1((0,1))$ and define an auxiliary function v such that

$$\phi(x)=v(x)\sin(\pi x). \tag{12}$$

Our goal is to show $I:=\int_0^1\left((\phi'(x))^2-\pi^2\phi^2(x)\right)dx\geq 0$.

1. Recall briefly why v is well-defined on the entire interval $(0,1)$.

   Solution : *The function* $\sin(\pi x)$ *is positive for all* $x\in(0,1)$*. Since* ϕ *has compact support in* $(0,1)$*, it is zero near the boundaries* $x=0$ *and* $x=1$*. Therefore, the function* $v(x)$ *defined by:*

   $$v(x)=\frac{\phi(x)}{\sin(\pi x)}$$

   is well-defined and continuously differentiable on the entire interval $(0,1)$.

2. Show that the full integrand can be written as:

   $$(\phi')^2-\pi^2\phi^2=(v')^2\sin^2(\pi x)+\frac{1}{2}\pi\left(v^2\right)'\sin(2\pi x)+\pi^2v^2\cos(2\pi x) \tag{13}$$

   and deduce an expression of I.
   Hint: Use trigonometric identities.

   Solution : *Differentiating* ϕ *and substituting the expressions for* ϕ *and its derivative* ϕ' *into the integrand* $(\phi')^2-\pi^2\phi^2$*. Expanding the square* $(\phi')^2$*, we have.*

   $$\begin{aligned}(\phi')^2&=\left(v'\sin(\pi x)+\pi v\cos(\pi x)\right)^2\\&=(v')^2\sin^2(\pi x)+2\pi vv'\sin(\pi x)\cos(\pi x)+\pi^2v^2\cos^2(\pi x),\\\pi^2\phi^2&=\pi^2v^2\sin^2(\pi x).\end{aligned}$$

   Subtracting gives:

   $$(\phi')^2-\pi^2\phi^2=(v')^2\sin^2(\pi x)+2\pi vv'\sin(\pi x)\cos(\pi x)+\pi^2v^2(\cos^2(\pi x)-\sin^2(\pi x)).$$

   Using the identity $\cos(2\theta)=\cos^2\theta-\sin^2\theta$ *with* $\theta=\pi x$*:*

   $$\cos^2(\pi x)-\sin^2(\pi x)=\cos(2\pi x).$$

So the expression becomes:

$$(\phi')^2 - \pi^2\phi^2 = (v')^2\sin^2(\pi x) + 2\pi vv'\sin(\pi x)\cos(\pi x) + \pi^2 v^2\cos(2\pi x).$$

Note that $2\pi vv' = \pi(v^2)'$. *Also,* $2\sin(\pi x)\cos(\pi x) = \sin(2\pi x)$. *Therefore:*

$$\begin{aligned} 2\pi vv'\sin(\pi x)\cos(\pi x) &= \pi(v^2)'\cdot\sin(\pi x)\cos(\pi x) \\ &= \frac{1}{2}\pi(v^2)'\cdot 2\sin(\pi x)\cos(\pi x) \\ &= \frac{1}{2}\pi(v^2)'\sin(2\pi x). \end{aligned}$$

The final form of the integrand is:

$$(\phi')^2 - \pi^2\phi^2 = (v')^2\sin^2(\pi x) + \frac{1}{2}\pi(v^2)'\sin(2\pi x) + \pi^2 v^2\cos(2\pi x),$$

and we deduce that

$$I = \int_0^1 (v')^2\sin^2(\pi x) + \frac{1}{2}\pi(v^2)'\sin(2\pi x) + \pi^2 v^2\cos(2\pi x)dx.$$

.

3. Show that, in fact,

$$I = \int_0^1 (v')^2\sin^2(\pi x)dx, \tag{14}$$

and deduce the desired Poincaré inequality (11).

Solution : *Let us compute* $\int_0^1 (v^2)'\sin(2\pi x)dx$, *we use integration by parts. Let:*

$$\begin{aligned} u &= \sin(2\pi x) && \implies du = 2\pi\cos(2\pi x)dx, \\ dv &= (v^2)'dx && \implies v = v^2. \end{aligned}$$

Then:

$$\int_0^1 (v^2)'\sin(2\pi x)dx = \Big[v^2\sin(2\pi x)\Big]_0^1 - \int_0^1 v^2\cdot 2\pi\cos(2\pi x)dx.$$

The boundary term is $\left[v^2(x)\sin(2\pi x)\right]_{x=0}^{x=1}$. *Since* ϕ *has compact support in* $(0,1)$, $\phi(x) = 0$ *(and thus* $v(x) = 0$*) for* x *near* 0 *and* 1. *Therefore:*

$$\lim_{x\to 0^+} v^2(x)\sin(2\pi x) = 0 \quad \textit{and} \quad \lim_{x\to 1^-} v^2(x)\sin(2\pi x) = 0.$$

So the boundary term evaluates to 0. *Substituting back, we get:*

$$\int_0^1 (v^2)'\sin(2\pi x)dx = -2\pi\int_0^1 v^2\cos(2\pi x)dx.$$

Now, compute the combination from the functional:

$$\begin{aligned} \frac{1}{2}\pi\int_0^1 (v^2)'\sin(2\pi x)dx + \pi^2\int_0^1 v^2\cos(2\pi x)dx &= \frac{1}{2}\pi\left(-2\pi\int_0^1 v^2\cos(2\pi x)dx\right) + \pi^2\int_0^1 v^2\cos(2\pi x)dx \\ &= -\pi^2\int_0^1 v^2\cos(2\pi x)dx + \pi^2\int_0^1 v^2\cos(2\pi x)dx \\ &= 0. \end{aligned}$$

We deduce that

$$I = \int_0^1 (v')^2\sin^2(\pi x)dx \geq 0 \tag{15}$$

which concludes the proof of the Poincaré inequality.

Exercise 4: Thermal phase shift in a wall – Heat equation in periodic regime *Context and position of the problem.* We seek to model the propagation of daily temperature variations inside a wall. The wall is treated as a homogeneous medium occupying the half-space $x \geq 0$, where x denotes the depth in the wall. The temperature $T(t,x)$ satisfies the heat equation

$$\frac{\partial T}{\partial t}(t,x) = \alpha \frac{\partial^2 T}{\partial x^2}(t,x), \qquad t > 0, \quad x > 0, \tag{16}$$

where $\alpha > 0$ is the thermal diffusivity of the material, expressed in $\mathrm{m}^2\,\mathrm{s}^{-1}$.

The temperature at the outer surface varies periodically according to

$$T(t,0) = T_m + A\cos(\omega t), \qquad \omega = \frac{2\pi}{\tau}, \tag{17}$$

where T_m is the mean temperature, $A > 0$ the amplitude of the oscillations and $\tau = 24\,\mathrm{h}$ their period. We are interested in the established periodic regime: the solution is periodic in time and remains bounded as $x \to +\infty$.

Part 1: A periodic solution to the heat equation

We set $u(t,x) = T(t,x) - T_m$ and we seek the harmonic regime in the form

$$u(t,x) = a(x)\cos(\omega t) + b(x)\sin(\omega t), \tag{18}$$

where a and b are two real functions.

1. Show that
$$\alpha a'' = \omega b, \qquad \alpha b'' = -\omega a.$$
What conditions do a and b satisfy at $x = 0$? *Hint: Substitute* (18) *into the heat equation* (16) *and identify the coefficients of* $\cos(\omega t)$ *and* $\sin(\omega t)$.

Solution : *We write*
$$u(t,x) = T(t,x) - T_m = a(x)\cos(\omega t) + b(x)\sin(\omega t).$$
Its useful derivatives are
$$\frac{\partial u}{\partial t} = -\omega a(x)\sin(\omega t) + \omega b(x)\cos(\omega t),$$
and
$$\frac{\partial^2 u}{\partial x^2} = a''(x)\cos(\omega t) + b''(x)\sin(\omega t).$$
The equation $\partial_t u = \alpha \partial_{xx} u$ *gives, after identifying the coefficients of the cosine and the sine,*
$$\boxed{\alpha a'' = \omega b, \qquad \alpha b'' = -\omega a}.$$
At $x = 0$, *the condition* $u(t,0) = A\cos(\omega t)$ *immediately implies*
$$\boxed{a(0) = A, \qquad b(0) = 0}.$$

We set
$$k = \sqrt{\frac{\omega}{2\alpha}}.$$

2. By eliminating b from the previous system, show that
$$a^{(4)} + 4k^4 a = 0. \tag{19}$$

Solution : *The first equation of the system gives*
$$b = \frac{\alpha}{\omega} a''.$$
Differentiating twice and using the second equation, we find
$$\frac{\alpha^2}{\omega} a^{(4)} = -\omega a,$$
that is,
$$a^{(4)} + \frac{\omega^2}{\alpha^2} a = 0.$$
Let $k = \sqrt{\omega/(2\alpha)}$. *Since* $4k^4 = \omega^2/\alpha^2$, *this equation becomes*
$$a^{(4)} + 4k^4 a = 0.$$

3. Verify that any bounded solution equation (19) can be written

$$a(x) = \mathrm{e}^{-kx}(C\cos(kx) + D\sin(kx)).$$

Hint: Show that the solution to (19) *is a vector space of dimension* 4. *Look for solutions of the form* $a(x) = \exp(rx)$.

Solution : *It is obvious to show that he solution to* (19) *is a vector space. It is of dimension* 4 *since the solution is completly determined by* $a(0), a'(0), a''(0)$ *and* $a'''(0)$. *We the aim at solving* $r^4 = -4k^4$. *A direct verification shows that the four functions*

$$\mathrm{e}^{kx}\cos(kx), \quad \mathrm{e}^{kx}\sin(kx), \quad \mathrm{e}^{-kx}\cos(kx), \quad \mathrm{e}^{-kx}\sin(kx)$$

are solutions of it. They form a basis of the space of solutions. The boundedness as $x \to +\infty$ *therefore imposes*

$$a(x) = \mathrm{e}^{-kx}(C\cos(kx) + D\sin(kx)).$$

4. Show that

$$T(t,x) = T_m + A\mathrm{e}^{-x/\delta}\cos\left(\omega t - \frac{x}{\delta}\right)$$

where

$$\delta = \sqrt{\frac{2\alpha}{\omega}}$$

is called the thermal penetration depth. Deduce that

$$\delta = \sqrt{\frac{\alpha\tau}{\pi}}.$$

Hint: Use the relation between a *and* b, *then the conditions imposed at* $x = 0$.

Solution : *Two differentiations give*

$$a''(x) = 2k^2\mathrm{e}^{-kx}(-D\cos(kx) + C\sin(kx)).$$

Since $2k^2 = \omega/\alpha$, *the relation* $b = (\alpha/\omega)a''$ *leads to*

$$b(x) = \mathrm{e}^{-kx}(-D\cos(kx) + C\sin(kx)).$$

The conditions $a(0) = A$ *and* $b(0) = 0$ *give* $C = A$ *and* $D = 0$. *Thus,*

$$a(x) = A\mathrm{e}^{-kx}\cos(kx), \qquad b(x) = A\mathrm{e}^{-kx}\sin(kx).$$

Returning to u, *then using the identity* $\cos p\cos q + \sin p\sin q = \cos(q-p)$, *we obtain*

$$\begin{aligned} u(t,x) &= A\mathrm{e}^{-kx}(\cos(kx)\cos(\omega t) + \sin(kx)\sin(\omega t)) \\ &= A\mathrm{e}^{-kx}\cos(\omega t - kx). \end{aligned}$$

Setting $\delta = 1/k = \sqrt{2\alpha/\omega}$, *it finally follows that*

$$\boxed{T(t,x) = T_m + A\mathrm{e}^{-x/\delta}\cos\left(\omega t - \frac{x}{\delta}\right)}.$$

Since $\omega = 2\pi/\tau$,

$$\delta = \sqrt{\frac{2\alpha}{\omega}} = \sqrt{\frac{2\alpha}{2\pi/\tau}} = \boxed{\sqrt{\frac{\alpha\tau}{\pi}}}.$$

Part 2: Thermal attenuation and phase shift

We consider a depth $L > 0$.

5. Show that the amplitude of the temperature oscillations at this depth equals

$$A_L = A\mathrm{e}^{-L/\delta},$$

and deduce the attenuation factor A_L/A.

Solution : *At depth L,*

$$T(t,L) = T_m + A\mathrm{e}^{-L/\delta}\cos\left(\omega t - \frac{L}{\delta}\right).$$

The amplitude and the attenuation factor are therefore respectively

$$\boxed{A_L = A\mathrm{e}^{-L/\delta}}, \qquad \boxed{\frac{A_L}{A} = \mathrm{e}^{-L/\delta}}.$$

6. How does A_L/A vary when the thickness L increases? When the thermal diffusivity α decreases? Interpret these results physically.

Solution : *When L increases, the factor $\mathrm{e}^{-L/\delta}$ decreases: temperature variations are attenuated more. Moreover, $\delta = \sqrt{2\alpha/\omega}$. If α decreases, δ decreases and L/δ increases; the attenuation factor therefore also decreases. A low diffusivity limits the penetration of thermal fluctuations.*

At the surface of the wall, a maximum of temperature is reached when

$$\omega t = 2k\pi, \qquad k \in \mathbb{Z}.$$

7. Show that at depth L, the same maximum is reached with a delay

$$\Delta t = \frac{L}{\delta\omega}.$$

Using the expression for δ, establish the formulas

$$\Delta t = L\sqrt{\frac{1}{2\alpha\omega}} = L\sqrt{\frac{\tau}{4\pi\alpha}}.$$

How does the phase shift vary when α decreases?

Solution : *At the surface, the maxima are reached when $\omega t = 2k\pi$. At depth L, they are reached when*

$$\omega t - \frac{L}{\delta} = 2k\pi.$$

The delay Δt is therefore defined by

$$\omega\Delta t = \frac{L}{\delta},$$

hence

$$\boxed{\Delta t = \frac{L}{\delta\omega}}.$$

Now $\delta\omega = \sqrt{2\alpha\omega}$. Consequently,

$$\boxed{\Delta t = L\sqrt{\frac{1}{2\alpha\omega}} = L\sqrt{\frac{\tau}{4\pi\alpha}}}.$$

In particular, $\Delta t \propto \alpha^{-1/2}$: the lower the diffusivity, the greater the phase shift.

Part 3: Comparison of materials and sizing

We consider a wall of thickness $L = 20\,\mathrm{cm}$ subjected to external variations of period $\tau = 24\,\mathrm{h}$. Two materials are considered:

dense calcium silicate block (Silka type): $\alpha_1 = 5{,}0 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$,

rigid wood fibre board (STEICOspecial dry type): $\alpha_2 = 1{,}5 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$.

These diffusivities are representative rounded values.

8. For each of the two materials, calculate the thermal penetration depth δ, the attenuation factor A_L/A and the time phase shift Δt expressed in hours.
Which material best meets the objective of 12 h? Compare the attenuations obtained.

Solution : *We first convert the period into seconds:*

$$\tau = 24 \times 3600 = 86\,400\,\text{s}.$$

For the dense calcium silicate block (Silka),

$$\delta_1 = \sqrt{\frac{\alpha_1 \tau}{\pi}} = \sqrt{\frac{5{,}0 \times 10^{-7} \times 86\,400}{\pi}} \simeq 0{,}117\,\text{m} = 11{,}7\,\text{cm}.$$

We deduce

$$\frac{A_{L,1}}{A} = \mathrm{e}^{-L/\delta_1} \simeq \mathrm{e}^{-1{,}71} \simeq 0{,}18,$$

and

$$\Delta t_1 = \frac{L}{\delta_1}\frac{\tau}{2\pi} \simeq 6{,}5\,\text{h}.$$

For the rigid wood fibre board (STEICOspecial dry), the same calculations give

$$\delta_2 = \sqrt{\frac{1{,}5 \times 10^{-7} \times 86\,400}{\pi}} \simeq 0{,}0642\,\text{m} = 6{,}42\,\text{cm},$$
$$\frac{A_{L,2}}{A} = \mathrm{e}^{-L/\delta_2} \simeq \mathrm{e}^{-3{,}11} \simeq 0{,}044,$$
$$\Delta t_2 = \frac{L}{\delta_2}\frac{\tau}{2\pi} \simeq 11{,}9\,\text{h}.$$

The results are summarized in the following table:

	δ	A_L/A	Δt
Silka	11,7 cm	0,18	6,5 h
STEICOspecial dry	6,42 cm	0,044	11,9 h

The manufacturer data lead to approximately $5{,}06 \times 10^{-7}\,\text{m}^2\,\text{s}^{-1}$ *for Silka and* $1{,}36$ *to* $1{,}43 \times 10^{-7}\,\text{m}^2\,\text{s}^{-1}$ *for STEICOspecial dry: the values in the statement are therefore reasonable rounded values.*

The wood fibre board is closer to the objective of 12 h *and attenuates more: at* 20 cm*, the remaining amplitude is about* 4,4 %*, against* 18 % *for Silka.*

Exercise 5 — Numerical solution of the one-dimensional heat equation We aim to solve numerically the Cauchy problem

$$\begin{cases} \frac{\partial u}{\partial t}(t,x) = \alpha \frac{\partial^2 u}{\partial^2 x}(t,x), & t > 0, \quad x \in (0,L), \\ u(0,x) = u_0(x), & x \in (0,L), \\ u(t,0) = u(t,L) = 0, & t > 0, \end{cases} \tag{20}$$

where $\alpha > 0$ and $L > 0$. Introduce the grid

$$\Delta x = \frac{L}{I}, \quad x_i = i\Delta x, \quad i = 0, \ldots, I, \tag{21}$$

$$\Delta t > 0, \quad t_n = n\Delta t, \quad n \in \mathbb{N}. \tag{22}$$

We denote $u_i^n \approx u(t_n, x_i)$ and we introduce the diffusion number:

$$\mu = \frac{\alpha \Delta t}{(\Delta x)^2}. \tag{23}$$

1. Approximate the second derivative $\dfrac{\partial^2 u}{\partial x^2}(t_n, x_i)$ by a second-order centered finite-difference approximation.

2. Write the explicit forward Euler scheme associated with the heat equation.

3. Write the scheme in the form
$$U^{n+1} = BU^n, \tag{24}$$
specifying the structure of the matrix B.

4. Show that stability in the L^2 norm of the explicit scheme requires a condition on μ, and explicit this condition.

5. Write pseudocode or a Python script implementing the explicit scheme.

6. Test numerically the explicit scheme with and without satisfying the CFL condition.

Implicit scheme

7. Write the implicit forward Euler scheme associated with the heat equation.

8. Write the scheme in the form
$$CU^{n+1} = U^n, \tag{25}$$
specifying the structure of the matrix C.

9. What is the main computational cost of advancing the solution by one time step with this method?

10. Performing a similar Von Neumann stability analysis to the one made for the explicit scheme, show that the implicit scheme is unconditionally stable.

11. Write pseudocode or a Python script implementing the implicit scheme.

12. Test numerically the implicit scheme to verify its unconditional stability.

Crank-Nicolson scheme

13. The Crank-Nicolson scheme is given by:
$$\frac{U_i^{n+1} - U_i^n}{\Delta t} = \frac{\alpha}{2}\left(\frac{U_{i-1}^{n+1} - 2U_i^{n+1} + U_{i+1}^{n+1}}{(\Delta x)^2} + \frac{U_{i-1}^n - 2U_i^n + U_{i+1}^n}{(\Delta x)^2}\right). \tag{26}$$

14. Conducting a Von Neumann stability analysis, show that the Crank-Nicolson scheme is unconditionally stable.

15. What is the order of the truncation error in time and space.

16. Implement the Crank-Nicolson scheme and compare its accuracy with both the explicit and implicit schemes.

References

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